Problem 4.35: Parity of the Neutrino & the τ–θ Puzzle

Griffiths, *Introduction to Elementary Particles*

A Detailed, Pedagogical Solution

Problem Statement (Griffiths 4.35)

(a) Is the neutrino an eigenstate of P? If so, what is its intrinsic parity?
(b) Now that we know τ^+ and θ^+ are actually both the K^+, which of the decays in Equation 4.53 actually violates parity conservation?

Equation 4.53:

$$\theta^+ \to \pi^+ + \pi^0 \qquad (P = +1)$$

$$\tau^+ \to \begin{cases} \pi^+ + \pi^0 + \pi^0 \\ \pi^+ + \pi^+ + \pi^- \end{cases} \qquad (P = -1)$$

1 Background: What is Parity?

Before we solve the problem, let's build up the required concepts step by step.

The **parity operator** P performs a spatial inversion, i.e. it reflects all spatial coordinates through the origin:

$$P: \quad \mathbf{r} \to -\mathbf{r}$$

Equivalently, $(x, y, z) \to (-x, -y, -z)$.
If a state $|\psi\rangle$ is an **eigenstate** of P, then:

$$P\,|\psi\rangle = \eta_P\,|\psi\rangle\,, \qquad \eta_P = \pm 1$$

The eigenvalue η_P is called the **intrinsic parity** of the particle.

1.1 Intrinsic Parity of Particles

Every particle can (in principle) be assigned an **intrinsic parity** $\eta_P = \pm 1$, provided that parity is conserved in the interactions that produce or destroy it.

- By **convention**, quarks (and leptons) are assigned intrinsic parity $+1$.
- **Antiquarks** (and antileptons) have the opposite parity -1 (a consequence of the Dirac equation for fermion–antifermion pairs).
- For a system of particles, parity is **multiplicative**:

$$P_{\text{total}} = \eta_{P_1} \cdot \eta_{P_2} \cdots \eta_{P_n} \cdot (-1)^L$$

  where L is the orbital angular momentum quantum number.
- Parity is conserved in the **strong** and **electromagnetic** interactions, but is **violated** in the **weak** interaction.

1.2 Intrinsic Parity of the Pion

Since this problem revolves around pion final states, let's recall:

The pion (π) is a pseudoscalar meson: it is a bound state of a quark and an antiquark in a state with $L = 0$. Since a quark has parity $+1$ and an antiquark has parity -1:

$$P_\pi = (+1)(-1)(-1)^0 = \boxed{-1}$$

All pions (π^+, π^-, π^0) have intrinsic parity -1.

2 Part (a): Is the Neutrino an Eigenstate of P?

This part asks a deep and subtle question. Let us reason through it carefully.

2.1 What Does it Mean to Be an Eigenstate of P?

For a particle to be an eigenstate of P, the parity operation must map the particle's state back onto itself (up to a sign). That is, $P|\nu\rangle = \pm|\nu\rangle$.

2.2 What Does P Do to a Neutrino?

The key observation is about **helicity**. The parity operator reverses momentum $\mathbf{p} \to -\mathbf{p}$ but does **not** change spin $\mathbf{S}$ (since spin is an axial vector). Therefore, parity **flips the helicity**:

$$P|\nu_L\rangle = \eta_P|\nu_R\rangle$$

where ν_L denotes a left-handed neutrino and ν_R denotes a right-handed neutrino. Parity converts a left-handed state into a right-handed state (and vice versa).

2.3 The Crucial Experimental Fact

Experimentally, **only left-handed neutrinos** and **right-handed antineutrinos** are observed in nature. In the Standard Model (with massless neutrinos), there are **no right-handed neutrinos** (ν_R).

This means:

$$P\,|\nu_L\rangle \propto |\nu_R\rangle$$

But $|\nu_R\rangle$ **does not exist** (or at least does not participate in any known interaction). Therefore, the result of applying P to the neutrino state does not give back the neutrino state—it gives a state that is not observed!

No, the neutrino is NOT an eigenstate of P.

The parity operator reverses momentum but not spin, so it converts a left-handed neutrino into a right-handed neutrino:

$$P\,|\nu_L\rangle = \eta_P\,|\nu_R\rangle$$

Since right-handed neutrinos do not exist in the Standard Model, $P\,|\nu_L\rangle$ is not proportional to $|\nu_L\rangle$. Therefore, the neutrino is not an eigenstate of P, and **it does not have a well-defined intrinsic parity.**

This is itself a manifestation of parity violation in the weak interaction: the weak force couples only to left-handed particles, making the very existence of the neutrino (which interacts only weakly) inherently parity-violating.

If right-handed neutrinos did exist and participated in interactions, then we could assign the neutrino an intrinsic parity (by convention, +1 for fermions). But since the weak interaction is the *only* interaction neutrinos participate in, and the weak interaction maximally violates parity, the concept of intrinsic parity simply does not apply to neutrinos in a meaningful way.

Note: With the discovery of neutrino oscillations (and hence neutrino masses), the situation becomes more nuanced. Massive neutrinos can in principle have both helicity states, but the right-handed component is suppressed by a factor of $(m_\nu/E)^2$, which is extraordinarily small.

3 Part (b): The τ–θ Puzzle and Parity Violation

3.1 Historical Context: The τ–θ Puzzle

In the 1950s, physicists observed two apparently different particles:

- The "θ^+" particle, which decayed into two pions

- The "τ^+" particle, which decayed into three pions

These two particles had the **same mass**, the **same lifetime**, and **spin-0**. The only difference was their decay modes—and therefore apparently their parity quantum numbers.

If parity were conserved, a single particle could not decay into final states with *different* parities. So physicists concluded that τ^+ and θ^+ must be *different* particles.

The resolution, proposed by **T.D. Lee** and **C.N. Yang** in 1956, was revolutionary: parity is **not conserved** in the weak interaction. The τ^+ and θ^+ are actually the **same particle**—the K^+ meson—and its different decay modes simply violate parity conservation.

3.2 Computing the Parity of Each Decay Mode

Since the K^+ has spin-0, the final-state pions must also have total angular momentum $J = 0$, which means the orbital angular momentum is $L = 0$ for the two-body decay. Let us compute the parity of each final state.

3.2.1 Decay 1: $\theta^+ \to \pi^+ + \pi^0$

For this two-body decay with $L = 0$:

$$P_{\text{final}} = P_{\pi^+} \cdot P_{\pi^0} \cdot (-1)^L = (-1)(-1)(-1)^0 = (+1)(1) = \boxed{+1}$$

Step by step:

1. Each pion has intrinsic parity -1.
2. With two pions: $(-1) \times (-1) = +1$.
3. The orbital angular momentum contribution: $(-1)^0 = +1$.
4. Total parity of the final state: $P = +1$.

3.2.2 Decay 2: $\tau^+ \to \pi^+ + \pi^0 + \pi^0$ (or $\pi^+ + \pi^+ + \pi^-$)

For the three-body decay, the situation is more involved. There are two relative orbital angular momenta (L_1 for one pair and L_2 for the third pion relative to that pair). However, since the parent K^+ has spin-0, we need $L_1 + L_2 = 0$, so the simplest (and dominant) configuration is $L_1 = L_2 = 0$.
With $L_1 = L_2 = 0$:

$$P_{\text{final}} = P_\pi \cdot P_\pi \cdot P_\pi \cdot (-1)^{L_1} \cdot (-1)^{L_2} = (-1)^3 \cdot (+1) \cdot (+1) = \boxed{-1}$$

Step by step:

1. Three pions, each with intrinsic parity -1: $(-1)^3 = -1$.

2. Orbital angular momentum contributions: $(-1)^0 \times (-1)^0 = +1$.

3. Total parity of the final state: $P = -1$.

This result holds for both $\pi^+\pi^0\pi^0$ and $\pi^+\pi^+\pi^-$ channels.

(R.H. Dalitz confirmed this using his famous Dalitz plot analysis in 1953–1954, establishing the spin-parity of the three-pion final state as 0^-.)

3.3 What is the Parity of the K^+?

Now we need to know the intrinsic parity of the K^+ meson itself.

The K^+ is a pseudoscalar meson composed of an up quark and a strange antiquark ($u\bar{s}$) in a state with $L = 0$:

$$P_{K^+} = P_u \cdot P_{\bar{s}} \cdot (-1)^L = (+1)(-1)(-1)^0 = \boxed{-1}$$

The K^+ has intrinsic parity -1.

3.4 Which Decay Violates Parity?

Now we can compare the parity of the initial state (K^+) with the parity of each final state:

Decay Mode	$\boldsymbol{P}_{\text{initial}}$ (K^+)	$\boldsymbol{P}_{\text{final}}$	**Parity Conserved?**
$K^+ \to \pi^+ + \pi^0$ (θ mode)	-1	$+1$	**NO**
$K^+ \to \pi^+\pi^0\pi^0$ (τ mode)	-1	-1	**YES**
$K^+ \to \pi^+\pi^+\pi^-$ (τ mode)	-1	-1	**YES**

The θ^+ **decay mode**, i.e.

$$K^+ \to \pi^+ + \pi^0$$

is the one that **violates parity conservation**.

The initial state K^+ has parity $P = -1$, but the two-pion final state has parity $P = +1$. Since $P_{\text{initial}} \neq P_{\text{final}}$, parity is not conserved in this decay.

The τ^+ decay modes ($K^+ \to 3\pi$) have final-state parity $P = -1$, which *matches* the parity of the K^+. These decays **conserve parity** (at least as far as the parity quantum number is concerned).

4 Summary and Key Takeaways

Part (a): The neutrino is **not an eigenstate of parity**. Parity flips helicity ($\nu_L \to \nu_R$), but right-handed neutrinos do not exist in the Standard Model. Therefore, the neutrino cannot have a well-defined intrinsic parity.

Part (b): Since the K^+ has intrinsic parity $P = -1$:

- The decay $K^+ \to \pi^+\pi^0$ (θ mode) produces a final state with $P = +1$. This **violates parity**.
- The decays $K^+ \to 3\pi$ (τ modes) produce final states with $P = -1$. These **conserve parity**.

The τ–θ puzzle was one of the most important problems in 20th-century physics. Its resolution by Lee and Yang in 1956—that the weak interaction violates parity—was confirmed experimentally by C.S. Wu in 1957 using polarized ^{60}Co nuclei. Lee and Yang received the Nobel Prize in Physics in 1957 for this groundbreaking insight.

The lesson: a symmetry that seems "obvious" or "beautiful" need not be a symmetry of nature. The laws of physics *do* distinguish between left and right.

5 Appendix: Quick Reference of Parity Rules

Rule	**Formula / Statement**
Parity of a system	$P_{\text{total}} = \prod_i \eta_{P_i} \cdot (-1)^L$
Fermion intrinsic parity	$+1$ (by convention)
Antifermion intrinsic parity	-1 (from Dirac equation)
Pseudoscalar mesons ($q\bar{q}$, $L = 0$)	$P = -1$
Scalar mesons ($q\bar{q}$, $L = 1$)	$P = +1$
Photon	$P = -1$ (vector)
n pions (all at $L = 0$)	$P = (-1)^n$
Parity conserved in	Strong & EM interactions
Parity violated in	Weak interaction